STM32F4神舟号_原理图_V2.0

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Page 6	以太网LAN8720A/音频WM8978

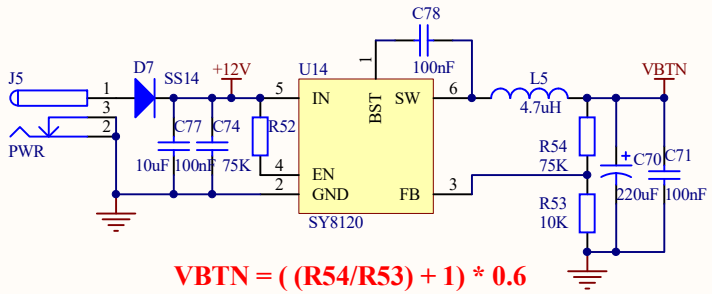
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Date: 2021-04-09	Sheet of	
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历史版本

版本号	日 期	描 述
V2.0	2020-8-10	初始版本

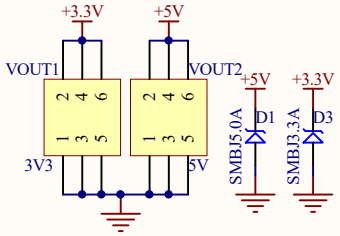
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DC IN: 6V~24V　　DC/DC电源稳压

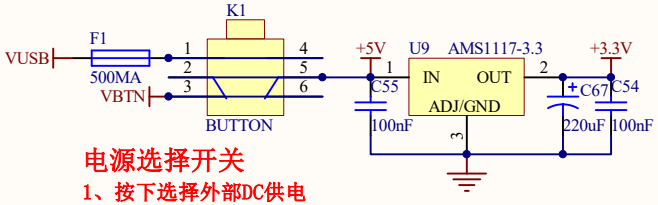


VBTN = ((R54/R53) + 1) * 0.6

3.3V　5V 电源输出



LDO　3.3V/1.8V

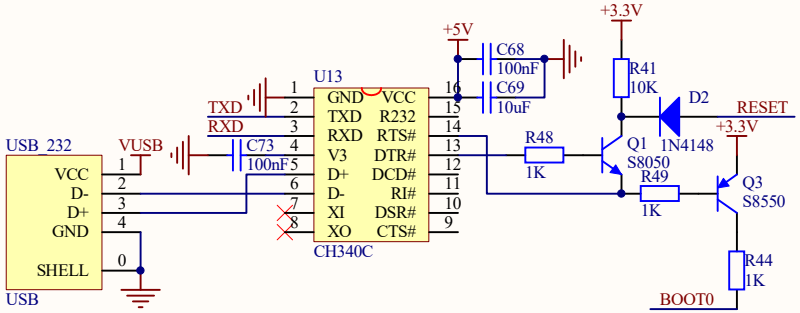


电源选择开关

1、按下选择外部DC供电

2、弹起选择USB供电

USB转串口模块　ISP下载



Title	STM32F4神舟号_原理图	郑州精畅科技
Size: A4	Number: 主电源/USB232	Revision: V2.0
Date: 2021-04-09	Sheet of	
File: E:\STM32F4神舟号原理图\..STM32F4神舟号_Draw.SchDoc		

引出GPIO口

GPIO引出110个，除了晶振占用4个IO其他全部引出

P9 Header 30X2

FSMC_A8	PF14	1 2	PF15	FSMC_A9
FSMC_A10	PG0	3 4	PG1	FSMC_A11
FSMC_A12	PG2	5 6	PG3	FSMC_A13
FSMC_A14	PG4	7 8	PG5	FSMC_A15
FSMC_A16	PD11	9 10	PE11	FSMC_D8
FSMC_D9	PE12	11 12	PE13	FSMC_D10
FSMC_D7	PE10	13 14	PE9	FSMC_D6
FSMC_D5	PE8	15 16	PE7	FSMC_D4
FSMC_A7	PF13	17 18	PF12	FSMC_A6
FSMC_D3	PD1	19 20	PD0	FSMC_D2
FSMC_D1	PD15	21 22	PD14	FSMC_D0
FSMC_NE3	PG10	23 24	PF0	FSMC_A0
FSMC_A1	PF1	25 26	PF2	FSMC_A2
FSMC_A3	PF3	27 28	PF4	FSMC_A4
USART3_RX	PB11	29 30	PB10	USART3_TX
	PD5	31 32	PF11	
BOOT1	PB2	33 34	PB1	
	PB0	35 36	PA3	USART2_RX
	PA4	37 38	PA2	USART2_TX
	PC0	39 40	PA0	WKUP
VDDA		41 42	VREF+	
	RESET	43 44	PF10	
	PF9	45 46	PC13	
VBAT		47 48	PE5	
	PE4	49 50	PE3	
	PE2	51 52	PF6	
	PA5	53 54	PE1	
	PE0	55 56		
+3.3V		57 58	+3.3V	
		59 60		

P8 Header 30X2

FSMC_A18	PD13	1 2	PD12	FSMC_A17
FSMC_D12	PE15	3 4	PE14	FSMC_D11
FSMC_D14	PD9	5 6	PD8	FSMC_D13
FSMC_A5	PF5	7 8	PD10	FSMC_D15
	PB12	9 10	PD4	
SPI2_MISO	PB14	11 12	PB13	SPI2_SCK
	PG12	13 14	PB15	LCD_BL
	PC4	15 16	PC5	
	PA1	17 18	PA7	SPI2_MOSI
	PG14	19 20	PC1	
	PG11	21 22	PG13	
	PC3	23 24	PD3	
	PC6	25 26	PC2	
SDIO_D0	PC8	27 28	PC7	
	PA8	29 30	PC9	SDIO_D1
USART1_RX	PA10	31 32	PA9	USART1_TX
	PA12	33 34	PA11	
JTCK	PA14	35 36	PA13	JTMS
SDIO_D2	PC10	37 38	PA15	JTDI
SDIO_CLK	PC12	39 40	PC11	SDIO_D3
	PD6	41 42	PD2	SDIO_CMD
	PG9	43 44	PD7	
JTDO	PB3	45 46	PG15	
JTRST	PB5	47 48	PB4	
	PB7	49 50	PB6	
	PB8	51 52	BOOT0	
	PA6	53 54	PB9	
	PE6	55 56	PF8	
	PG8	57 58	PF7	
	PG7	59 60	PG6	

USB_UART/USART1

P7 USART1

PA10_USART1_RX	2 1	PA9_USART1_TX
TXD	4 3	RXD

JTAG

JTAG

R2	JTRST	1	VDD	VDD	2
R3 10KJTDI		3	TRST	GND	4
R5 10KJTMS		5	TDI	GND	6
10KJTCK		7	TMS/SWDIO	GND	8
		9	TCK/SWCLK	GND	10
		11	NC	GND	12
R6	JTDO	13	TDO/SWO	GND	14
R4 10K	10KRESET	15	RESET#	GND	16
		17	NC	GND	18
		19	NC	GND	20

+3.3V

MCU最小系统

U10 STM32F407ZGT6

	WK_UP	PA0	34	PA0-WKUP/U2_CTS/U4_TX/ETH_MII_CRS/TIM2_CH1_ETR/TIM5_CH1/TIM8_ETR/ADC123_IN0	PE0/TIM4_ETR/FSMC_NBL0/DCMI_D2
RMII_REF_CLK		PA1	35	PA1/U2_RTS/U4_RX/ETH_MII_REF_CLK/ETH_MII_RX_CLK/TIM2_CH2/TIM5_CH2/ADC123_IN1	PE1/FSMC_NBL1/DCMI_D3
ETH_MDIO	USART2_TX	PA2	36	PA2/U2_TX/TIM2_CH3/TIM5_CH3/TIM9_CH1/ETH_MDIO/ADC123_IN2	PE2/TRACECLK/FSMC_A23/ETH_MII_TXD3
PWM_DAC	USART2_RX	PA3	37	PA3/U2_RX/TIM2_CH4/TIM5_CH4/TIM9_CH2/OTG_HS_ULPI_D0/ETH_MII_COL/ADC123_IN3	PE3/TRACED0/FSMC_A19
STM_DAC	DCMI_HREF	PA4	40	PA4/SPI1_NSS/U2_CK/DCMI_HSYNC/OTG_HS_SOF/I2S3_WS/DAC1_OUT/ADC12_IN4	PE4/TRACED1/FSMC_A20/DCMI_D4
	STM_ADC	PA5	41	PA5/SPI1_SCK/OTG_HS_ULPI_CK/TIM2_CH1_ETR/TIM8_CH1N/DAC2_OUT/ADC12_IN5	PE5/TRACED2/FSMC_A21/TIM9_CH1/DCMI_D6
	DCMI_PCLK	PA6	42	PA6/SPI1_MISO/TIM1_BKIN/TIM3_CH1/TIM8_BKIN/TIM13_CH1/DCMI_PIXCLK/ADC12_IN6	PE6/TRACED3/FSMC_A22/TIM9_CH2/DCMI_D7
RMII_CRS_DV		PA7	43	PA7/SPI1_MOSI/TIM1_CH1N/TIM3_CH2/TIM8_CH1N/TIM14_CH1/ETH_MII_RX_DV/ETH_RMII_CRS_DV/ADC12_IN7	PE7/FSMC_D4/TIM1_ETR
REMOTE_IN	DCMI_XCLK	PA8	100	PA8/TIM1_CH1/U1_CK/I2C3_SCL/MCO1/OTG_FS_SOF	PE8/FSMC_D5/TIM1_CH1N
	USART1_TX	PA9	101	PA9/TIM1_CH2/U1_TX/I2C3_SMBA/OTG_FS_VBUS/DCMI_D0	PE9/FSMC_D6/TIM1_CH1
	USART1_RX	PA10	102	PA10/TIM1_CH3/U1_RX/OTG_FS_ID/DCMI_D1	PE10/FSMC_D7/TIM1_CH2N
	USB_D-	PA11	103	PA11/TIM1_CH4/U1_CTS/CAN1_RX/OTG_FS_DM	PE11/FSMC_D8/TIM1_CH2
	USB_D+	PA12	104	PA12/TIM1_ETR/U1_RTS/CAN1_TX/OTG_FS_DP	PE12/FSMC_D9/TIM1_CH3N
	JTMS	PA13	105	PA13/JTMS_SWDIO	PE13/FSMC_D10/TIM1_CH3
	JTCK	PA14	109	PA14/JTCK_SWCLK	PE14/FSMC_D11/TIM1_CH4
	JTDI	PA15	110	PA15/JTDI/SPI1_NSS/SPI3_NSS/TIM2_CH1_ETR/I2S3_WS	PE15/FSMC_D12/TIM1_BKIN
	T_SCK	PB0	46	PB0/TIM1_CH2N/TIM3_CH3/TIM8_CH2N/OTG_HS_ULPI_D1/ETH_MII_RXD2/ADC12_IN8	PF0/FSMC_A0/I2C2_SDA
	T_PEN	PB1	47	PB1/TIM1_CH3N/TIM3_CH4/TIM8_CH3N/OTG_HS_INTN/ETH_MII_RXD3/ADC12_IN9	PF1/FSMC_A1/I2C2_SCL
T_MISO	BOOT1	PB2	48	PB2/BOOT1	PF2/FSMC_A2/I2C2_SMBA
SPI1_SCK	JTDO	PB3	133	PB3/JTDO/TRACESWO/TIM2_CH2/SPI1_SCK/SPI3_SCK/I2S3_CK	PF3/FSMC_A3/ADC3_IN9
SPI1_MISO	JTRST	PB4	134	PB4/JNTRST/TIM3_CH1/SPI1_MISO/SPI3_MISO/I2S3ext_SD	PF4/FSMC_A4/ADC3_IN14
SPI1_MOSI		PB5	135	PB5/TIM3_CH2/SPI1_MOSI/SPI3_MOSI/CAN2_RX/I2C1_SMBA/OTG_HS_ULPI_D7/ETH_PPS_OUT/DCMI_D10/I2S3_SD	PF5/FSMC_A5/ADC3_IN15
	DCMI_D5	PB6	136	PB6/TIM4_CH1/U1_TX/CAN2_TX/I2C1_SCL/DCMI_D5	PF6/TIM10_CH1/FSMC_NIORD/ADC3_IN4
DCMI_VSYNC		PB7	137	PB7/TIM4_CH2/U1_RX/FSMC_NL/I2C1_SDA/DCMI_VSYNC	PF7/TIM11_CH1/FSMC_NREG/ADC3_IN5
	IIC_SCL	PB8	139	PB8/TIM4_CH3/TIM10_CH1/CAN1_RX/SDIO_D4/ETH_MII_TXD3/I2C1_SCL/DCMI_D6	PF8/TIM13_CH1/FSMC_NIOWR/ADC3_IN6
	IIC_SDA	PB9	140	PB9/TIM4_CH4/TIM11_CH1/CAN1_TX/SDIO_D5/SPI2_NSS/I2C1_SDA/DCMI_D7/I2S2_WS	PF9/TIM14_CH1/FSMC_CD/ADC3_IN7
	USART3_TX	PB10	69	PB10/SPI2_SCK/TIM2_CH3/U3_TX/I2C2_SCL/OTG_HS_ULPI_D3/I2S2_CK	PF10/FSMC_INTR/ADC3_IN8
	USART3_RX	PB11	70	PB11/TIM2_CH4/U3_RX/I2C2_SDA/HS_ULPI_D4/ETH_MII_TX_EN/ETH_RMII_TX_EN	PF11/DCMI_D12
I2S_LRCK		PB12	73	PB12/SPI2_NSS/TIM1_BKIN/U3_CK/CAN2_RX/I2C2_SMBA/OTG_HS_ULPI_D5/OTG_HS_ID/ETH_MII_TXD0/ETH_RMII_TXD0/I2S2_WS	PF12/FSMC_A6
I2S_SCLK		PB13	74	PB13/SPI2_SCK/TIM1_CH1N/U3_CTS/CAN2_TX/OTG_HS_ULPI_D6/OTG_HS_VBUS/ETH_MII_TXD1/ETH_RMII_TXD1/I2S2_CK	PF13/FSMC_A7
	F_CS	PB14	75	PB14/SPI2_MISO/TIM1_CH2N/TIM8_CH2N/TIM12_CH1/U3_RTS/OTG_HS_DM/I2S2ext_SD	PF14/FSMC_A8
	LCD_BL	PB15	76	PB15/SPI2_MOSI/TIM1_CH3N/TIM8_CH3N/TIM12_CH2/OTG_HS_DP/I2S2_SD	PF15/FSMC_A9
GBC_LED		PC0	26	PC0/OTG_HS_ULPI_STP/ADC123_IN10	PG0/FSMC_A10
ETH_MDC		PC1	27	PC1/ETH_MDC/ADC123_IN11	PG1/FSMC_A11
I2S_SDOUT		PC2	28	PC2/SPI2_MISO/OTG_HS_ULPI_DIR/ETH_MII_TXD2/I2S2ext_SD/ADC123_IN12	PG2/FSMC_A12
I2S_SDIN		PC3	29	PC3/SPI2_MOSI/OTG_HS_ULPI_NXT/ETH_MII_TX_CLK/I2S2_SD/ADC123_IN13	PG3/FSMC_A13
RMII_RXD0		PC4	44	PC4/ETH_MII_RXD0/ETH_RMII_RXD0/ADC12_IN14	PG4/FSMC_A14
RMII_RXD1		PC5	45	PC5/ETH_MII_RXD1/ETH_RMII_RXD1/ADC12_IN15	PG5/FSMC_A15
I2S_MCLK	DCMI_D0	PC6	96	PC6/TIM3_CH1/TIM8_CH1/U6_TX/SDIO_D6/DCMI_D0/I2S2_MCK	PG6/FSMC_INT2
	DCMI_D1	PC7	97	PC7/TIM3_CH2/TIM8_CH2/U6_RX/SDIO_D7/DCMI_D1/I2S3_MCK	PG7/FSMC_INT3/U6_CK
SDIO_D0	DCMI_D2	PC8	98	PC8/TIM3_CH3/TIM8_CH3/U6_CK/SDIO_D0/DCMI_D2	PG8/U6_RTS/ETH_PPS_OUT
SDIO_D1	DCMI_D3	PC9	99	PC9/TIM3_CH4/TIM8_CH4/I2C3_SDA/SDIO_D1/DCMI_D3/MCO2/I2S_CKIN	PG9/FSMC_NE2/FSMC_NCE3/U6_RX
SDIO_D2		PC10	111	PC10/SPI3_SCK/U3_TX/U4_TX/SDIO_D2/DCMI_D8/I2S3_CK	PG10/FSMC_NCE4_1/FSMC_NE3
SDIO_D3	DCMI_D4	PC11	112	PC11/SPI3_MISO/U3_RX/U4_RX/SDIO_D3/DCMI_D4/I2S3ext_SD	PG11/FSMC_NCE4_2/ETH_MII_TX_EN/ETH_RMII_TX_EN
SDIO_SCK		PC12	113	PC12/SPI3_MOSI/U3_CK/U5_TX/SDIO_CK/DCMI_D9/I2S3_SD	PG12/FSMC_NE4/U6_RTS
	T_CS	PC13	7	PC13/RTC_AF1	PG13/FSMC_A24/U6_CTS/ETH_MII_TXD0/ETH_RMII_TXD0
		PC14	8	PC14-OSC32_IN	PG14/FSMC_A25/U6_TX/ETH_MII_TXD1/ETH_RMII_TXD1
		PC15	9	PC15-OSC32_OUT	PG15/U6_CTS/DCMI_D13
FSMC_D2		PD0	114	PD0/FSMC_D2/CAN1_RX	PH0-OSC_IN
FSMC_D3		PD1	115	PD1/FSMC_D3/CAN1_TX	PH1-OSC_OUT
SDIO_CMD		PD2	116	PD2/TIM3_ETR/U5_RX/SDIO_CMD/DCMI_D11	NRST
ETH_RESET		PD3	117	PD3/FSMC_CLK/U2_CTS	VBAT
FSMC_NOE		PD4	118	PD4/FSMC_NOE/U2_RTS	Vref+
FSMC_NWE		PD5	119	PD5/FSMC_NWE/U2_TX	Vcap_1
DCMI_SCL		PD6	122	PD6/FSMC_NWAIT/U2_RX	Vcap_2
DCMI_SDA		PD7	123	PD7/FSMC_NE1/FSMC_NCE2/U2_CK	VDDA
FSMC_D13		PD8	77	PD8/FSMC_D13/U3_TX	VSSA
FSMC_D14		PD9	78	PD9/FSMC_D14/U3_RX	PDR_ON
FSMC_D15		PD10	79	PD10/FSMC_D15/U3_CK	
FSMC_A16		PD11	80	PD11/FSMC_A16/FSMC_CLE/U3_CTS	
FSMC_A17		PD12	81	PD12/FSMC_A17/FSMC_ALE/TIM4_CH1/U3_RTS	
FSMC_A18		PD13	82	PD13/FSMC_A18/TIM4_CH2	
FSMC_D0		PD14	85	PD14/FSMC_D0/TIM4_CH3	
FSMC_D1		PD15	86	PD15/FSMC_D1/TIM4_CH4	
BOOT0			138	BOOT0	

32.768KHz Y3 C66 10pF C65 10pF

VSS ×8 (16 38 51 61 83 94 97 120 130) VDD ×10 (30 17 52 39 62 72 84 95 108 121 131 144)

+3.3V L3 GZ1608D601TF 磁珠600Ω@100MHz C63 C60 C59 C53 C45 C61 C31 C46 C36 C48 100nF ×10

VDD R43 10K VBAT D5 D4 1N4148 ×2 BAT1 PDR_ON R47 NC

R38 1M Y2 8MHz C62 22pF C64 22pF

RESET C32 2.2uF C50 2.2uF VDDA L4 GZ1608D601TF 磁珠600Ω@100MHz +3.3V C56 10uF C58 100nF

SRAM

U5 RAM IS62WV51216BLL-55TLI

FSMC_A0	5	A0	I/O15	38	FSMC_D15
FSMC_A1	4	A1	I/O14	37	FSMC_D14
FSMC_A2	3	A2	I/O13	36	FSMC_D13
FSMC_A3	2	A3	I/O12	35	FSMC_D12
FSMC_A4	1	A4	I/O11	32	FSMC_D11
FSMC_A5	44	A5	I/O10	31	FSMC_D10
FSMC_A6	43	A6	I/O9	30	FSMC_D9
FSMC_A7	42	A7	I/O8	29	FSMC_D8
FSMC_A8	27	A8	I/O7	16	FSMC_D7
FSMC_A9	26	A9	I/O6	15	FSMC_D6
FSMC_A10	25	A10	I/O5	14	FSMC_D5
FSMC_A11	24	A11	I/O4	13	FSMC_D4
FSMC_A12	22	A12	I/O3	10	FSMC_D3
FSMC_A13	21	A13	I/O2	9	FSMC_D2
FSMC_A14	20	A14	I/O1	8	FSMC_D1
FSMC_A15	19	A15	I/O0	7	FSMC_D0
FSMC_A16	18	A16			
FSMC_A17	23	A17			
FSMC_A18	28	A18			
		UB		40	FSMC_NBL1
FMC_VDD	11	VDD		39	FSMC_NBL0
	33	VDD		41	FSMC_NOE
	12	GND		17	FSMC_NWE
	34	GND		6	FSMC_NE3
		CE			

VDD R25 10K

+3.3V L2 GZ1608D601TF 磁珠600Ω@100MHz FMC_VDD C15 10uF C12 2.2uF C16 100nF C13 100nF

LCD液晶屏接口

P15 TFT_LCD

FSMC_NE4	1	LCD_CS	RS	2	FSMC_A6
FSMC_NWE	3	WR/CLK	RD	4	FSMC_NOE
RESET	5	RST	DB1	6	FSMC_D0
FSMC_D1	7	DB2	DB3	8	FSMC_D2
FSMC_D3	9	DB4	DB5	10	FSMC_D4
FSMC_D5	11	DB6	DB7	12	FSMC_D6
FSMC_D7	13	DB8	DB10	14	FSMC_D8
FSMC_D9	15	DB11	DB12	16	FSMC_D10
FSMC_D11	17	DB13	DB14	18	FSMC_D12
FSMC_D13	19	DB15	DB16	20	FSMC_D14
FSMC_D15	21	DB17	GND	22	
LCD_BL	23	BL	VDD3.3	24	
	25	VDD3.3	GND	26	
T_MISO	29	GND	BL_VDD	28	
T_PEN	31	MISO	MOSI	30	T_MOSI
T_CS	33	T_CS	CLK	32	T_SCK

+3.3V C1 100nF +5V

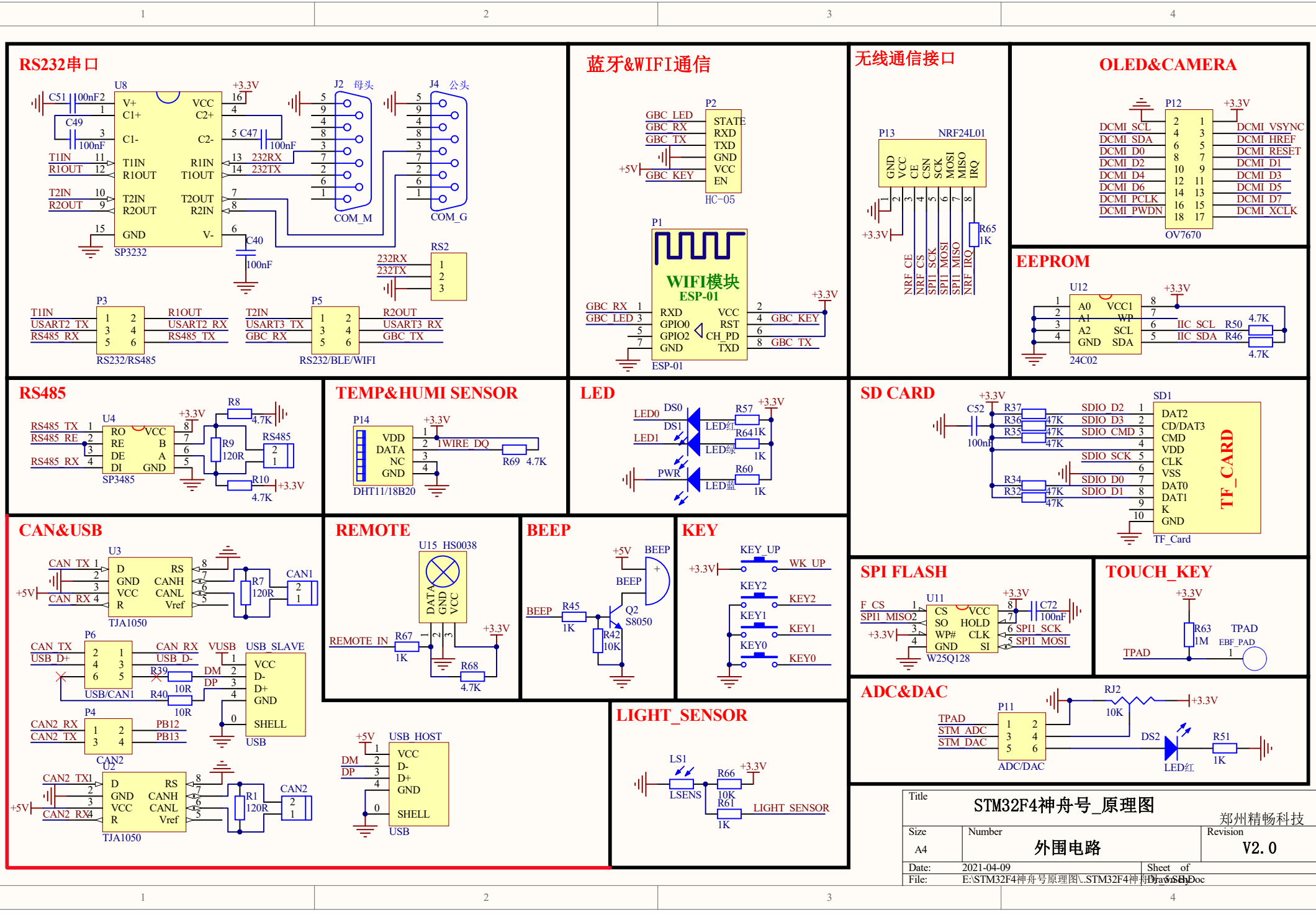
RESET复位键

+3.3V R62 10K RESET C79 100nF

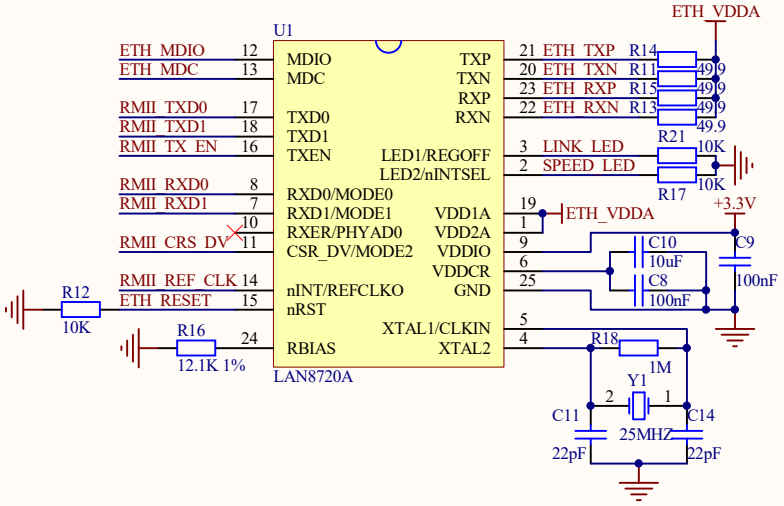
BOOT启动设置

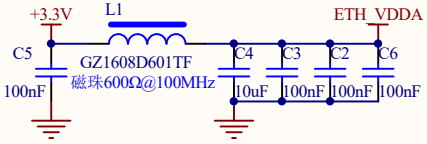
+3.3V BOOT R55 10K BOOT0 R56 10K BOOT1 Header 3X2

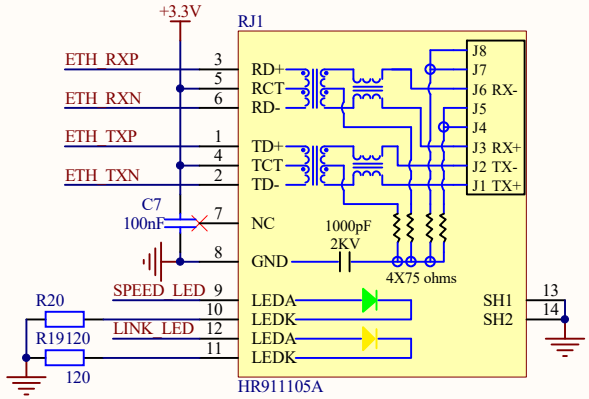
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Size	A2	Number MCU最小系统	Revision V2.0
Date:	2021-04-09	Sheet of	
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RS232串口
U8
SP3232
J2 母头
J4 公头
COM_M
COM_G
P3
RS232/RS485
P5
RS232/BLE/WIFI
RS2
蓝牙&WIFI通信
P2
HC-05
P1
WIFI模块
ESP-01
无线通信接口
P13
NRF24L01
OLED&CAMERA
P12
OV7670
EEPROM
U12
24C02
RS485
U4
SP3485
TEMP&HUMI SENSOR
P14
DHT11/18B20
LED
SD CARD
SD1
TF_Card
CAN&USB
U3
TJA1050
P6
USB/CAN1
P4
CAN2
U2
TJA1050
USB
REMOTE
U15 HS0038
BEEP
KEY
SPI FLASH
U11
W25Q128
TOUCH_KEY
ADC&DAC
P11
ADC/DAC
LIGHT_SENSOR
Title
STM32F4神舟号_原理图
郑州精畅科技
Size
A4
Number
外围电路
Revision
V2.0
Date:
2021-04-09
File:
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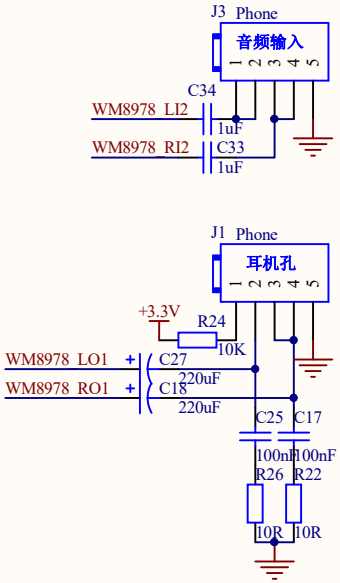
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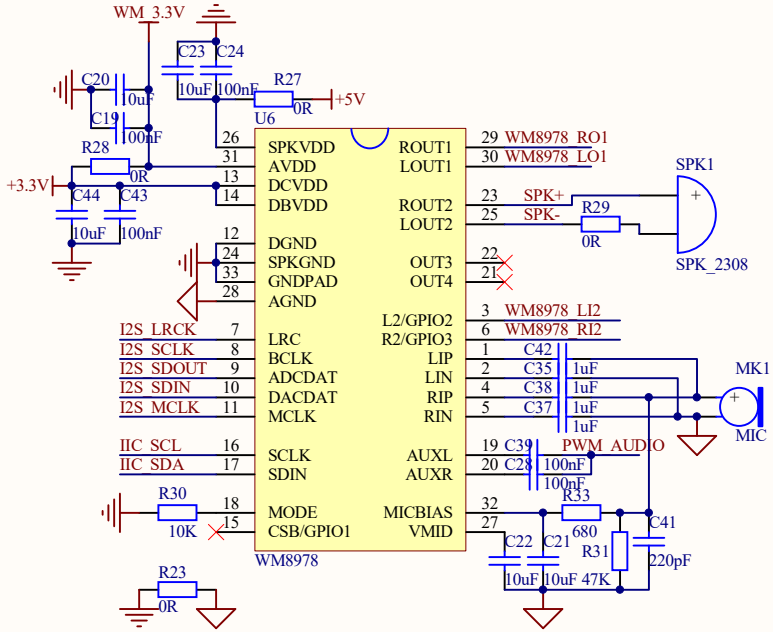




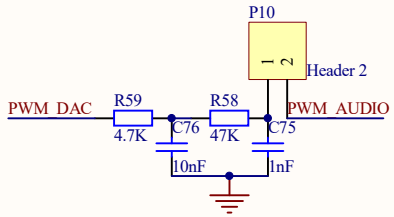


音频WM8978





PWM DAC/AUDIO



Title	STM32F4神舟号_原理图	郑州精畅科技
Size: A4	Number: 以太网/音频	Revision: V1.0
Date: 2021-04-09	Sheet of	
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